Outputs:

```
PS C:\Users\91779\OneDrive\Desktop\GU\DCST_system> javac *.java
PS C:\Users\91779\OneDrive\Desktop\GU\DCST_system> java Main

==================================
 DCST NIRF INFORMATION SYSTEM
          GOA UNIVERSITY
==================================
1. Student Details
2. Staff Details
3. Academic Activities
4. Non-Academic Activities
5. Courses and Subjects
6. Object Referencing and Cloning
7. Collections and Placements
8. Exception Handling
9. Reflection and Access Modifiers
10. Static Member Count
11. Save and Read NIRF Report
12. Generic Class and Method
13. OOP Demonstrations
0. Exit

Enter your choice: 1

===== STUDENT DETAILS =====

Student ID: 101
Name: Aarav
Email: aarav@gmail.com
Course: MCA
Academic Year: 2026
Marks: 85.0
Department: Computer Science and Technology

Student ID: 102
Name: Ananya
Email: ananya@gmail.com
Course: MCA
Academic Year: 2026
Marks: 90.0
Department: Computer Science and Technology
```

```
================================
 DCST NIRF INFORMATION SYSTEM
          GOA UNIVERSITY
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0. Exit

Enter your choice: 2

===== STAFF DETAILS =====

Teaching Staff
ID: 201
Name: Professor A
Designation: Assistant Professor
Subject: Java
Salary: 65000.0

Non-Teaching Staff
ID: 202
Name: Staff B
Designation: Office Assistant
Department: Administration
Salary: 30000.0

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 DCST NIRF INFORMATION SYSTEM
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```

```
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0. Exit

Enter your choice: 3

===== ACADEMIC ACTIVITIES =====
Activity: ISA
Type: ACADEMIC
Year: 2026
--------------------
Activity: Midpoint
Type: ACADEMIC
Year: 2026
--------------------
Activity: SEA
Type: ACADEMIC
Year: 2026
--------------------
Activity: Study Tour
Type: ACADEMIC
Year: 2026
--------------------
Other examples: Industrial Visit, Internship, Placement, Workshop, Seminar and Hackathon.

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```

```
0. Exit

Enter your choice: 4

===== NON-ACADEMIC ACTIVITIES =====
Activity: InfoFest
Type: NON_ACADEMIC
Year: 2026
--------------------
Activity: Plateaunia
Type: NON_ACADEMIC
Year: 2026
--------------------
Activity: Play-on
Type: NON_ACADEMIC
Year: 2026
--------------------

Unique Activity Names:
InfoFest
Plateaunia
Play-on

================================
 DCST NIRF INFORMATION SYSTEM
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0. Exit

Enter your choice: 5

===== PROGRAMS AND COURSES =====
MCA
Ph.D. Computer Science
```

```
Enter your choice: 5

===== PROGRAMS AND COURSES =====
MCA
Ph.D. Computer Science

Course: MCA
Duration: 2 years
Subjects:
MCA101 - Object Oriented Programming (4 credits)
MCA102 - Data Structures (4 credits)
MCA103 - Computer Networks (3 credits)

================================
 DCST NIRF INFORMATION SYSTEM
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0. Exit

Enter your choice: 6

===== OBJECT REFERENCING AND CLONING =====
Both students share the same Department object.
Same reference: true
Location through Student 2: Updated DCST Building

Original Department:
Department: Computer Science and Technology
University: Goa University
Location: New Campus Location

Shallow Clone:
Department: Computer Science and Technology
University: Goa University
Location: New Campus Location
```

Problems Output Debug Console Terminal Ports

```
Shallow Clone:
Department: Computer Science and Technology
University: Goa University
Location: New Campus Location

Deep Clone:
Department: Computer Science and Technology
University: Goa University
Location: Updated DCST Building

================================
 DCST NIRF INFORMATION SYSTEM
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0. Exit

Enter your choice: 7

===== COLLECTIONS =====

LIST - Students:
Aarav
Ananya

SET - Unique Activities:
InfoFest
Plateaunia
Play-on

MAP - Courses:
1 : MCA
2 : Ph.D. Computer Science

QUEUE - Placement:
Student 101
Student 102

Processing Placement Queue:
Student 101
Student 102

================================
 DCST NIRF INFORMATION SYSTEM
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0. Exit

Enter your choice: 8
Enter student marks: 9
Valid marks: 9.0
Marks validation completed.

================================
 DCST NIRF INFORMATION SYSTEM
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```

```
11. Save and Read NIRF Report
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0. Exit

Enter your choice: 10

===== STATIC COUNTERS =====
Total members created: 4
Total activities: 7

================================
 DCST NIRF INFORMATION SYSTEM
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0. Exit

Enter your choice: 11

===== FILE PERSISTENCE =====
Data saved successfully.
Data saved successfully.
Data saved successfully.

Saved DCST Information:
DCST NIRF Report - Goa University
Student: Aarav
Student: Ananya

================================
 DCST NIRF INFORMATION SYSTEM
         GOA UNIVERSITY
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0. Exit

Enter your choice: 12

===== GENERICS =====
Stored value: DCST Goa University
Stored value: 100

Generic Method:
Aarav
Ananya
DCST

================================
 DCST NIRF INFORMATION SYSTEM
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0. Exit
```

```
Enter your choice: 13

===== OOP DEMONSTRATIONS =====
Printing DCST information...
Generating NIRF report...
Technical Workshop conducted by DCST.

Student ID: 101
Name: Aarav
Email: aarav@gmail.com
Course: MCA
Academic Year: 2026
Marks: 90.0
Department: Computer Science and Technology

Teaching Staff
ID: 201
Name: Professor A
Designation: Assistant Professor
Subject: Java
Salary: 65000.0

================================
 DCST NIRF INFORMATION SYSTEM
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10. Static Member Count
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0. Exit

Enter your choice: 0
Thank you for using DCST NIRF System!
PS C:\Users\91779\OneDrive\Desktop\GU\DCST_system>
```